

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Friday 22 May 2026

1.1 Daily Limit

Limit Involving Dilogarithm and Logarithmic Functions

Evaluate the limit:

$$\lim_{x \rightarrow \infty} \left(-\text{Li}_2(-x) - \frac{1}{2} \ln^2(x+1) \right)$$

Solution: To evaluate the limit as $x \rightarrow \infty$, we utilize Abel's inversion formula for the dilogarithm function $\text{Li}_2(z)$, which states that for $z < 0$:

$$\text{Li}_2(z) + \text{Li}_2\left(\frac{1}{z}\right) = -\frac{\pi^2}{6} - \frac{1}{2} \ln^2(-z)$$

Letting $z = -x$ where $x > 0$, we obtain:

$$\text{Li}_2(-x) + \text{Li}_2\left(-\frac{1}{x}\right) = -\frac{\pi^2}{6} - \frac{1}{2} \ln^2(x)$$

Rearranging this equation gives the asymptotic expansion for $-\text{Li}_2(-x)$:

$$-\text{Li}_2(-x) = \frac{\pi^2}{6} + \frac{1}{2} \ln^2(x) + \text{Li}_2\left(-\frac{1}{x}\right)$$

Now, substitute this expansion back into the original limit expression:

$$L = \lim_{x \rightarrow \infty} \left(\frac{\pi^2}{6} + \frac{1}{2} \ln^2(x) + \text{Li}_2\left(-\frac{1}{x}\right) - \frac{1}{2} \ln^2(x+1) \right)$$

We group the logarithmic terms together and observe their behavior:

$$\begin{aligned} \frac{1}{2} \ln^2(x) - \frac{1}{2} \ln^2(x+1) &= \frac{1}{2} (\ln(x) - \ln(x+1)) (\ln(x) + \ln(x+1)) \\ &= -\frac{1}{2} \ln\left(1 + \frac{1}{x}\right) \ln(x^2 + x) \end{aligned}$$

As $x \rightarrow \infty$, we can use the Maclaurin approximation $\ln\left(1 + \frac{1}{x}\right) \approx \frac{1}{x}$:

$$\lim_{x \rightarrow \infty} \left[-\frac{1}{2} \left(\frac{1}{x} - O\left(\frac{1}{x^2}\right) \right) \left(2 \ln(x) + \ln\left(1 + \frac{1}{x}\right) \right) \right] = \lim_{x \rightarrow \infty} \left(-\frac{\ln(x)}{x} \right) = 0$$

Furthermore, as $x \rightarrow \infty$, $-\frac{1}{x} \rightarrow 0$, so by the continuity of the dilogarithm function at the origin:

$$\lim_{x \rightarrow \infty} \text{Li}_2\left(-\frac{1}{x}\right) = \text{Li}_2(0) = 0$$

Since both the difference of the squared logarithms and the dilogarithm remainder vanish as $x \rightarrow \infty$, the limit evaluates strictly to the constant term:

$$\frac{\pi^2}{6}$$

Saturday 23 May 2026

2.1 Daily Limit

Limit of an Infinite Product with Cosines

Evaluate the limit:

$$\lim_{x \rightarrow 0^+} \left(\prod_{n=1}^{\infty} \cos \left(\frac{x}{2^n} \right) \right)^{\frac{1}{x^2}}$$

Solution: First, we simplify the infinite product inside the base using a standard trigonometric identity. Consider the finite partial product:

$$P_N(x) = \prod_{n=1}^N \cos \left(\frac{x}{2^n} \right) = \cos \left(\frac{x}{2} \right) \cos \left(\frac{x}{4} \right) \cos \left(\frac{x}{8} \right) \cdots \cos \left(\frac{x}{2^N} \right)$$

Multiplying and dividing by $2^N \sin \left(\frac{x}{2^N} \right)$ and applying the double-angle formula $\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$ repeatedly:

$$P_N(x) = \frac{2^N \sin \left(\frac{x}{2^N} \right) \cos \left(\frac{x}{2^N} \right) \cos \left(\frac{x}{2^{N-1}} \right) \cdots \cos \left(\frac{x}{2} \right)}{2^N \sin \left(\frac{x}{2^N} \right)} = \frac{\sin(x)}{2^N \sin \left(\frac{x}{2^N} \right)}$$

Taking the limit as $N \rightarrow \infty$, notice that $2^N \sin \left(\frac{x}{2^N} \right) = x \cdot \frac{\sin(x/2^N)}{x/2^N} \rightarrow x \cdot 1 = x$. Therefore:

$$\prod_{n=1}^{\infty} \cos \left(\frac{x}{2^n} \right) = \frac{\sin(x)}{x}$$

The original limit expression simplifies to:

$$L = \lim_{x \rightarrow 0^+} \left(\frac{\sin(x)}{x} \right)^{\frac{1}{x^2}}$$

To evaluate this indeterminate form 1^∞ , we take the natural logarithm of both sides:

$$\ln L = \lim_{x \rightarrow 0^+} \frac{1}{x^2} \ln \left(\frac{\sin(x)}{x} \right)$$

Using the Maclaurin series expansion of $\sin(x)$ around $x = 0$:

$$\frac{\sin(x)}{x} = \frac{x - \frac{x^3}{6} + \frac{x^5}{120} - O(x^7)}{x} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - O(x^6)$$

Substituting this into the Taylor expansion for $\ln(1 - u) \approx -u - \frac{u^2}{2}$:

$$\ln \left(\frac{\sin(x)}{x} \right) = \ln \left(1 - \left(\frac{x^2}{6} - \frac{x^4}{120} \right) \right) = - \left(\frac{x^2}{6} - \frac{x^4}{120} \right) - O(x^4) = -\frac{x^2}{6} + O(x^4)$$

Dividing by x^2 and taking the limit:

$$\ln L = \lim_{x \rightarrow 0^+} \frac{-\frac{x^2}{6} + O(x^4)}{x^2} = -\frac{1}{6}$$

Exponentiating both sides yields the final limit:

$$L = e^{-1/6}$$

Sunday 24 May 2026

3.1 Daily Limit

Limit of a Weighted Sum of Cosines

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \left[\frac{\pi}{2n} \sum_{k=1}^n \cos \left(\frac{\pi(n-k)}{2n} \right) \right]$$

Solution: We begin by algebraic simplification of the argument within the cosine function:

$$\frac{\pi(n-k)}{2n} = \frac{\pi}{2} - \frac{\pi k}{2n}$$

Using the complementary angle identity $\cos \left(\frac{\pi}{2} - \theta \right) = \sin(\theta)$, the sum transforms into:

$$S_n = \frac{\pi}{2n} \sum_{k=1}^n \sin \left(\frac{\pi k}{2n} \right)$$

We can factor out the constant $\frac{\pi}{2}$ to clearly expose the structure of a definite integral:

$$S_n = \frac{\pi}{2} \left[\frac{1}{n} \sum_{k=1}^n \sin \left(\frac{\pi}{2} \cdot \frac{k}{n} \right) \right]$$

Notice that the expression inside the brackets is a right-endpoint Riemann sum for the continuous function $f(x) = \sin \left(\frac{\pi}{2} x \right)$ over the interval $[0, 1]$, where the partition width is $\Delta x = \frac{1}{n}$ and the sample points are $x_k = \frac{k}{n}$.

As $n \rightarrow \infty$, the Riemann sum converges exactly to the definite integral:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \sin \left(\frac{\pi}{2} x_k \right) = \int_0^1 \sin \left(\frac{\pi}{2} x \right) dx$$

Evaluating this definite integral using standard antiderivatives:

$$\int_0^1 \sin \left(\frac{\pi}{2} x \right) dx = \left[-\frac{2}{\pi} \cos \left(\frac{\pi}{2} x \right) \right]_0^1 = -\frac{2}{\pi} \cos \left(\frac{\pi}{2} \right) - \left(-\frac{2}{\pi} \cos(0) \right) = 0 + \frac{2}{\pi} = \frac{2}{\pi}$$

Multiplying by the leading constant factor $\frac{\pi}{2}$ that was factored out initially:

$$\lim_{n \rightarrow \infty} S_n = \frac{\pi}{2} \cdot \frac{2}{\pi} = 1$$

Monday 25 May 2026

4.1 Daily Limit

Infinite Series with a Cubic Polynomial Denominator

Evaluate the limit:

$$\lim_{N \rightarrow \infty} \sum_{r=1}^N \frac{1}{r^2(r+1)(r+2)}$$

Solution: To evaluate this infinite series, we perform a partial fraction decomposition on the general term $a_r = \frac{1}{r^2(r+1)(r+2)}$. We seek constants A, B, C, D such that:

$$\frac{1}{r^2(r+1)(r+2)} = \frac{A}{r} + \frac{B}{r^2} + \frac{C}{r+1} + \frac{D}{r+2}$$

Multiplying through by the common denominator $r^2(r+1)(r+2)$ yields the identity:

$$1 = Ar(r+1)(r+2) + B(r+1)(r+2) + Cr^2(r+2) + Dr^2(r+1)$$

We determine the coefficients by substituting convenient roots: 1. Setting $r = 0$: $1 = B(1)(2) \implies B = \frac{1}{2}$. 2. Setting $r = -1$: $1 = C(-1)^2(1) \implies C = 1$. 3. Setting $r = -2$: $1 = D(-2)^2(-1) \implies D = -\frac{1}{4}$. 4. Equating coefficients of r^3 : $0 = A + C + D \implies A = -1 - (-\frac{1}{4}) = -\frac{3}{4}$.

Thus, the decomposed general term is:

$$a_r = \frac{1}{2r^2} - \frac{3}{4r} + \frac{1}{r+1} - \frac{1}{4(r+2)}$$

We split the infinite sum into two distinct convergent parts:

$$\sum_{r=1}^{\infty} a_r = \frac{1}{2} \sum_{r=1}^{\infty} \frac{1}{r^2} + \sum_{r=1}^{\infty} \left(-\frac{3}{4r} + \frac{1}{r+1} - \frac{1}{4(r+2)} \right)$$

The first sum is the renowned Basel problem:

$$\frac{1}{2} \sum_{r=1}^{\infty} \frac{1}{r^2} = \frac{1}{2} \left(\frac{\pi^2}{6} \right) = \frac{\pi^2}{12}$$

For the second sum, we algebraically regroup the rational terms to reveal standard telescoping series:

$$-\frac{3}{4r} + \frac{1}{r+1} - \frac{1}{4(r+2)} = -\frac{3}{4} \left(\frac{1}{r} - \frac{1}{r+1} \right) + \frac{1}{4} \left(\frac{1}{r+1} - \frac{1}{r+2} \right)$$

Summing both telescoping components from $r = 1$ to ∞ :

$$\begin{aligned} \sum_{r=1}^{\infty} \left(\frac{1}{r} - \frac{1}{r+1} \right) &= \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \cdots = 1 \\ \sum_{r=1}^{\infty} \left(\frac{1}{r+1} - \frac{1}{r+2} \right) &= \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \cdots = \frac{1}{2} \end{aligned}$$

Combining the telescoping sums gives:

$$-\frac{3}{4}(1) + \frac{1}{4} \left(\frac{1}{2} \right) = -\frac{3}{4} + \frac{1}{8} = -\frac{5}{8}$$

Adding both components together, the total limit evaluates to:

$$\frac{\pi^2}{12} - \frac{5}{8}$$

Tuesday 26 May 2026

5.1 Daily Limit

Limit of Exponential Expressions

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \frac{2^{n+1} + 3^{n+1}}{2^n + 3^n}$$

Solution: In limits involving sums of exponential terms as $n \rightarrow \infty$, the behavior is entirely governed by the base with the largest absolute value. Here, 3^n grows strictly faster than 2^n .

To evaluate the limit rigorously without indeterminate forms, we factor out the dominant power 3^{n+1} from the numerator and 3^n from the denominator:

$$\frac{2^{n+1} + 3^{n+1}}{2^n + 3^n} = \frac{3^{n+1} \left(\frac{2^{n+1}}{3^{n+1}} + 1 \right)}{3^n \left(\frac{2^n}{3^n} + 1 \right)}$$

Using exponent rules, we simplify the ratio of the factored terms:

$$\frac{3^{n+1}}{3^n} \cdot \frac{\left(\frac{2}{3}\right)^{n+1} + 1}{\left(\frac{2}{3}\right)^n + 1} = 3 \cdot \frac{\left(\frac{2}{3}\right)^{n+1} + 1}{\left(\frac{2}{3}\right)^n + 1}$$

Since $\left|\frac{2}{3}\right| < 1$, taking the limit as $n \rightarrow \infty$ causes the geometric sequences $\left(\frac{2}{3}\right)^n$ and $\left(\frac{2}{3}\right)^{n+1}$ to converge strictly to zero:

$$\lim_{n \rightarrow \infty} \left(\frac{2}{3}\right)^n = 0, \quad \lim_{n \rightarrow \infty} \left(\frac{2}{3}\right)^{n+1} = 0$$

Substituting these zero limits into our factored expression yields:

$$3 \cdot \frac{0 + 1}{0 + 1} = 3 \cdot 1 = 3$$

Wednesday 27 May 2026

6.1 Daily Limit

Limit of Gaussian Integral Moments

If $\lim_{T \rightarrow \infty} \int_0^T e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$, then evaluate:

$$\lim_{T \rightarrow \infty} \int_0^T x^2 e^{-x^2} dx$$

Solution: We define the integral of interest as a function of the upper limit T :

$$I(T) = \int_0^T x^2 e^{-x^2} dx$$

We apply the method of integration by parts. Let us decompose the integrand by setting:

$$u = x \implies du = dx$$

$$dv = x e^{-x^2} dx \implies v = \int x e^{-x^2} dx = -\frac{1}{2} e^{-x^2}$$

Applying the integration by parts formula $\int_0^T u dv = [uv]_0^T - \int_0^T v du$:

$$I(T) = \left[-\frac{1}{2} x e^{-x^2} \right]_0^T - \int_0^T \left(-\frac{1}{2} e^{-x^2} \right) dx$$

Evaluating the boundary term at the limits 0 and T :

$$\left[-\frac{1}{2} x e^{-x^2} \right]_0^T = -\frac{1}{2} T e^{-T^2} - (0) = -\frac{T}{2e^{T^2}}$$

Thus, the integral becomes:

$$I(T) = -\frac{T}{2e^{T^2}} + \frac{1}{2} \int_0^T e^{-x^2} dx$$

Now, we take the limit as $T \rightarrow \infty$ for both components on the right-hand side: 1. For the algebraic-exponential fraction, exponential growth dominates polynomial growth. By L'Hôpital's Rule:

$$\lim_{T \rightarrow \infty} \frac{T}{2e^{T^2}} = \lim_{T \rightarrow \infty} \frac{1}{4Te^{T^2}} = 0$$

2. For the integral term, we use the given Gaussian integral value:

$$\lim_{T \rightarrow \infty} \frac{1}{2} \int_0^T e^{-x^2} dx = \frac{1}{2} \left(\frac{\sqrt{\pi}}{2} \right) = \frac{\sqrt{\pi}}{4}$$

Summing the two limits gives the final result:

$$\lim_{T \rightarrow \infty} \int_0^T x^2 e^{-x^2} dx = 0 + \frac{\sqrt{\pi}}{4} = \frac{\sqrt{\pi}}{4}$$

Thursday 28 May 2026

7.1 Daily Limit

Infinite Product Limit Involving Cubes

Evaluate the infinite product limit:

$$\lim_{n \rightarrow \infty} \prod_{r=3}^n \left(\frac{r^3 - 8}{r^3 + 8} \right)$$

Solution: To evaluate this infinite product, we factor both the numerator and denominator using the algebraic identities for the difference and sum of cubes:

$$r^3 - 8 = (r - 2)(r^2 + 2r + 4)$$

$$r^3 + 8 = (r + 2)(r^2 - 2r + 4)$$

Let us define the quadratic polynomial $f(r) = r^2 - 2r + 4$. Observe what happens when we evaluate $f(r + 2)$:

$$f(r + 2) = (r + 2)^2 - 2(r + 2) + 4 = (r^2 + 4r + 4) - (2r + 4) + 4 = r^2 + 2r + 4$$

This establishes a crucial shift identity: the quadratic factor in the numerator is exactly $f(r + 2)$, while the quadratic factor in the denominator is $f(r)$. The finite product P_n can therefore be factored into two separate products:

$$P_n = \left[\prod_{r=3}^n \frac{r - 2}{r + 2} \right] \cdot \left[\prod_{r=3}^n \frac{f(r + 2)}{f(r)} \right]$$

Let us analyze the first product (the linear terms) by writing out its initial and final factors:

$$\prod_{r=3}^n \frac{r - 2}{r + 2} = \left(\frac{1}{5} \right) \left(\frac{2}{6} \right) \left(\frac{3}{7} \right) \left(\frac{4}{8} \right) \cdots \left(\frac{n - 4}{n} \right) \left(\frac{n - 3}{n + 1} \right) \left(\frac{n - 2}{n + 2} \right)$$

Due to telescoping cancellation, all terms from 5 to $n - 2$ cancel completely, leaving only the first four numerators and the last four denominators:

$$\prod_{r=3}^n \frac{r - 2}{r + 2} = \frac{1 \cdot 2 \cdot 3 \cdot 4}{(n - 1)n(n + 1)(n + 2)} = \frac{24}{(n - 1)n(n + 1)(n + 2)}$$

Next, let us analyze the second product (the quadratic terms):

$$\prod_{r=3}^n \frac{f(r + 2)}{f(r)} = \frac{f(5)}{f(3)} \cdot \frac{f(6)}{f(4)} \cdot \frac{f(7)}{f(5)} \cdot \frac{f(8)}{f(6)} \cdots \frac{f(n + 1)}{f(n - 1)} \cdot \frac{f(n + 2)}{f(n)}$$

Here, all intermediate terms $f(5), f(6), \dots, f(n)$ cancel out across a shift of two indices, leaving:

$$\prod_{r=3}^n \frac{f(r + 2)}{f(r)} = \frac{f(n + 1)f(n + 2)}{f(3)f(4)}$$

Evaluating the constants at the lower bound:

$$f(3) = 3^2 - 2(3) + 4 = 7, \quad f(4) = 4^2 - 2(4) + 4 = 12 \implies f(3)f(4) = 84$$

Combining both simplified products into our expression for P_n :

$$P_n = \frac{24}{84} \cdot \frac{f(n+1)f(n+2)}{(n-1)n(n+1)(n+2)}$$

As $n \rightarrow \infty$, both $f(n+1)$ and $f(n+2)$ behave asymptotically as n^2 , meaning their product grows as $n^4 + O(n^3)$. The denominator is a monic quartic polynomial $(n-1)n(n+1)(n+2) = n^4 + O(n^3)$. Thus, the degree of n is identical in top and bottom:

$$\lim_{n \rightarrow \infty} \frac{f(n+1)f(n+2)}{(n-1)n(n+1)(n+2)} = 1$$

Therefore, the limit evaluates cleanly to the simplified fraction:

$$\frac{24}{84} = \frac{2}{7}$$

Friday 29 May 2026

8.1 Daily Limit

Limit Involving Trigonometric and Radical Terms

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\sin^2(x)}{\sqrt{2} - \sqrt{1 + \cos(x)}}$$

Solution: Direct substitution of $x = 0$ yields the indeterminate form $\frac{0}{\sqrt{2} - \sqrt{2}} = \frac{0}{0}$. To resolve this, we rationalize the denominator by multiplying both numerator and denominator by the conjugate expression $(\sqrt{2} + \sqrt{1 + \cos(x)})$:

$$L = \lim_{x \rightarrow 0} \frac{\sin^2(x) (\sqrt{2} + \sqrt{1 + \cos(x)})}{(\sqrt{2} - \sqrt{1 + \cos(x)}) (\sqrt{2} + \sqrt{1 + \cos(x)})}$$

Applying the difference of squares formula $(a - b)(a + b) = a^2 - b^2$ to the denominator:

$$(\sqrt{2})^2 - (\sqrt{1 + \cos(x)})^2 = 2 - (1 + \cos(x)) = 1 - \cos(x)$$

The limit expression simplifies to:

$$L = \lim_{x \rightarrow 0} \frac{\sin^2(x) (\sqrt{2} + \sqrt{1 + \cos(x)})}{1 - \cos(x)}$$

We now apply the fundamental Pythagorean trigonometric identity $\sin^2(x) = 1 - \cos^2(x)$, which factors as $(1 - \cos(x))(1 + \cos(x))$:

$$L = \lim_{x \rightarrow 0} \frac{(1 - \cos(x))(1 + \cos(x)) (\sqrt{2} + \sqrt{1 + \cos(x)})}{1 - \cos(x)}$$

We cancel the common non-zero factor $(1 - \cos(x))$ for all $x \neq 0$ in the neighborhood of the origin:

$$L = \lim_{x \rightarrow 0} \left[(1 + \cos(x)) (\sqrt{2} + \sqrt{1 + \cos(x)}) \right]$$

Now that the indeterminacy is completely removed, we evaluate the limit by direct substitution of $x = 0$, noting that $\cos(0) = 1$:

$$L = (1 + 1) (\sqrt{2} + \sqrt{1 + 1}) = 2 (\sqrt{2} + \sqrt{2}) = 2(2\sqrt{2}) = 4\sqrt{2}$$

Saturday 30 May 2026

9.1 Daily Limit

Limit Involving Factorial (Gamma) Functions

Evaluate the limit:

$$\lim_{x \rightarrow 0} \left(\frac{x! + x}{x! - x} \right)^{\frac{1}{x}}$$

Solution: For real numbers, the factorial function $x!$ is formally defined via the Gamma function as $x! = \Gamma(x + 1)$. To understand its behavior near $x = 0$, we utilize the Taylor series expansion of $\Gamma(x + 1)$ around the origin:

$$x! = \Gamma(x + 1) = 1 - \gamma x + \frac{1}{2} \left(\gamma^2 + \frac{\pi^2}{6} \right) x^2 + O(x^3)$$

where $\gamma \approx 0.5772156649$ is the Euler-Mascheroni constant.

Let us substitute this expansion into the fraction inside the base of our limit:

$$\frac{x! + x}{x! - x} = \frac{(1 - \gamma x + O(x^2)) + x}{(1 - \gamma x + O(x^2)) - x} = \frac{1 + (1 - \gamma)x + O(x^2)}{1 - (1 + \gamma)x + O(x^2)}$$

We simplify this rational function using the geometric series expansion $\frac{1}{1-u} = 1 + u + O(u^2)$ where $u = (1 + \gamma)x + O(x^2)$:

$$\frac{x! + x}{x! - x} = [1 + (1 - \gamma)x + O(x^2)] [1 + (1 + \gamma)x + O(x^2)]$$

Multiplying the linear terms and collecting coefficients of x :

$$\frac{x! + x}{x! - x} = 1 + [(1 - \gamma) + (1 + \gamma)]x + O(x^2) = 1 + 2x + O(x^2)$$

Now, substitute this simplified base back into the original limit:

$$L = \lim_{x \rightarrow 0} (1 + 2x + O(x^2))^{\frac{1}{x}}$$

We recognize this as the standard exponential definition limit of the form $\lim_{u \rightarrow 0} (1 + ku)^{1/u} = e^k$. Rigorously, by taking the natural logarithm:

$$\ln L = \lim_{x \rightarrow 0} \frac{1}{x} \ln (1 + 2x + O(x^2)) = \lim_{x \rightarrow 0} \frac{2x + O(x^2)}{x} = 2$$

Exponentiating both sides yields the final value:

$$L = e^2$$

Sunday 31 May 2026

10.1 Daily Limit

Limit Involving Harmonic Numbers

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \frac{n^2}{H_n \sum_{k=1}^n \frac{k}{H_k}}$$

where $H_n = \sum_{j=1}^n \frac{1}{j}$ represents the n -th harmonic number.

Solution: Let us denote the sequence in the denominator sum as $A_n = \sum_{k=1}^n \frac{k}{H_k}$ and define $B_n = \frac{n^2}{H_n}$. We seek to evaluate the limit $L = \lim_{n \rightarrow \infty} \frac{B_n}{A_n}$. We invert the ratio and apply the Stolz-Cesàro theorem to evaluate $\lim_{n \rightarrow \infty} \frac{A_n}{B_n}$:

$$\lim_{n \rightarrow \infty} \frac{A_n}{B_n} = \lim_{n \rightarrow \infty} \frac{A_n - A_{n-1}}{B_n - B_{n-1}}$$

First, the forward difference of the numerator sequence A_n is simply its n -th term:

$$A_n - A_{n-1} = \frac{n}{H_n}$$

Next, we calculate the forward difference of the denominator sequence B_n :

$$B_n - B_{n-1} = \frac{n^2}{H_n} - \frac{(n-1)^2}{H_{n-1}}$$

Recall the recurrence relation for harmonic numbers: $H_{n-1} = H_n - \frac{1}{n}$. Substituting this into the second term:

$$\frac{(n-1)^2}{H_{n-1}} = \frac{n^2 - 2n + 1}{H_n \left(1 - \frac{1}{nH_n}\right)}$$

For large n , we expand the denominator using the asymptotic binomial series $\frac{1}{1-\varepsilon} = 1 + \varepsilon + O(\varepsilon^2)$ where $\varepsilon = \frac{1}{nH_n}$:

$$\begin{aligned} \frac{(n-1)^2}{H_{n-1}} &= \frac{n^2 - 2n + 1}{H_n} \left(1 + \frac{1}{nH_n} + O\left(\frac{1}{n^2 H_n^2}\right)\right) \\ &= \frac{n^2 - 2n + 1}{H_n} + \frac{n^2}{H_n} \left(\frac{1}{nH_n}\right) - O\left(\frac{1}{H_n}\right) = \frac{n^2 - 2n + 1}{H_n} + \frac{n}{H_n^2} - O\left(\frac{1}{H_n}\right) \end{aligned}$$

Subtracting this expression from $\frac{n^2}{H_n}$ yields:

$$B_n - B_{n-1} = \frac{n^2}{H_n} - \left[\frac{n^2 - 2n + 1}{H_n} + \frac{n}{H_n^2} - O\left(\frac{1}{H_n}\right) \right] = \frac{2n-1}{H_n} - \frac{n}{H_n^2} + O\left(\frac{1}{H_n}\right)$$

Now, substitute both differences back into our Stolz-Cesàro ratio:

$$\lim_{n \rightarrow \infty} \frac{A_n - A_{n-1}}{B_n - B_{n-1}} = \lim_{n \rightarrow \infty} \frac{\frac{n}{H_n}}{\frac{2n-1}{H_n} - \frac{n}{H_n^2} + O\left(\frac{1}{H_n}\right)}$$

We multiply both numerator and denominator by $\frac{H_n}{n}$ to simplify the expression:

$$= \lim_{n \rightarrow \infty} \frac{1}{\left(2 - \frac{1}{n}\right) - \frac{1}{H_n} + O\left(\frac{1}{n}\right)}$$

Since $H_n \sim \ln(n) \rightarrow \infty$ as $n \rightarrow \infty$, terms such as $\frac{1}{n}$ and $\frac{1}{H_n}$ vanish completely:

$$\lim_{n \rightarrow \infty} \frac{A_n}{B_n} = \frac{1}{2 - 0 - 0} = \frac{1}{2}$$

Inverting this result back to our original limit expression gives:

$$\lim_{n \rightarrow \infty} \frac{n^2}{H_n \sum_{k=1}^n \frac{k}{H_k}} = \frac{1}{1/2} = 2$$